BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 230 · Programming paradigms: applied comparison

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does OOP differ from a procedural program?

Q2. What does a local variable mean?

Q3. If a field is private-set, how can callers change it safely?

Q4. Why does `Deposit(balance,5);` not update the caller's int balance if only a return value is produced?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

refactoring — Restructuring code while preserving expected behaviour.

encapsulation — Controlled access to internal state.

instance — Individual object created from a class.

return value — Output returned to the caller.

side effect — Observable change beyond a returned result.

New learning

Procedural state — The caller passes an integer balance and explicitly stores the returned value.

Object state — Account object A stores Balance internally and Deposit changes that object's state.

Separate instances — Account B stays at 20 when Account A changes to 15; the objects have independent state.

Worked example

Two accounts start at 8 and 20; only first account deposits 4. Trace both.

1. Predict — Account a's Balance becomes 12.

2. Model response — Account b remains at 20 because its state is separate.

We Do · Practise with guidance

A procedural function returns balance+3, but the caller ignores the return. What happens?

```
int balance=7;
Deposit(balance,3); // result unused
// What is balance?
```

Predict

Deposit computes and returns 10, but its result is ignored.

Model response

The caller's balance remains 7. Assign balance=Deposit(balance,3) to store 10.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Procedural AddFive(x) returns x+5. If caller x=2 discards the result, what is x?

Question 2

Two objects have balances A=10, B=30. A.Deposit(7) and B.Deposit(2) run. Find both balances and describe why they differ.

Question 3

Compare one maintenance advantage and one risk of procedural and OOP implementations for a library system.

Exit Ticket · Answer before checking

Calling a C# function that returns a new balance automatically changes the caller's original int variable.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. OOP groups state and behaviour into objects; procedural code is organised around procedures/functions.

Q2. A variable accessible only within its declaration's scope.

Q3. Through public methods/property logic defined by the class.

Q4. The return is discarded and a value-type parameter is a copy; the caller must assign the returned result.

You Do pages · Independent questions

Q1. [3] Function returns 7 [1]; caller does not assign it [1]; x remains 2 [1].

Q2. [6] A=17 [1], B=32 [1]; each is a separate instance [2], each method changes only its own state [2].

Q3. [6] Procedures can be individually reused/tested [1] but shared mutable data can create coupling [2]; OOP encapsulates per-book state [1] and validation in methods [1], but excessive class complexity can make a small solution harder to follow [1].

Exit Ticket · Model answer

False. The caller must assign the returned value unless code uses a ref parameter or other shared state.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding